Failure Analysis and Data Patching in Small-Scale Language Models

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https://github.com/jeffZ23/stat359.git
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Introduction

Small language models trained on narrow-domain corpora present a tractable experimental setting for studying failure mode structure: their limited capacity makes systematic breakdowns interpretable, and the compactness of their training distribution allows targeted interventions. TinyStoriesLLM, a family of transformer-based models trained on the TinyStories corpus of simple English narratives (Eldan & Li, 2023), is an illustrative case. Despite producing fluent surface text, models at this scale routinely exhibit recurring qualitative failures- degenerate repetition, narrative self-contradiction, and loss of character identity- that reduce outputs to unusable or misleading text.

The central question motivating this work is whether targeted fine-tuning on small, curated datasets can suppress specific, identifiable failure modes without access to large-scale supervision or model retraining. Two model variants are evaluated: a base completion model and an instruction-tuned chat model derived from the same backbone. The thesis is that failure mode severity and patchability differ fundamentally between these two variants due to an asymmetry between capacity-bounded architectural limitations (base model) and mislabeled behavioral alignment (chat model), and that as few as 40 curated training examples can produce measurable behavioral change in the latter.

Background

Transformer language models autoregressively model the conditional distribution $p(x_t \mid x_{<t})$ over discrete token sequences. At inference time, a next-token distribution is computed as:

$$p(x_t \mid x_{<t}) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

where Q, K, V are the query, key, and value projections derived from the current context. Models operating with local attention windows of width w can only attend to tokens within the interval

$[t - w, t - 1]$; when $w = 256$ (as in the base model evaluated here), referents introduced more than 256 tokens prior are invisible to the current computation, imposing a structural constraint on long-range coreference.

Small language models are known to exhibit characteristic failure modes at the boundary of their representational capacity. Repetition loops arise when the model assigns high probability to a recently generated phrase, reinforcing its own output in a feedback cycle; this is related to the exposure bias problem noted in sequence-to-sequence literature. Coherence failures occur when locally plausible continuations contradict prior assertions that fall outside the model's effective context window. Character confusion is a specific instance of the broader coreference resolution problem, which demands persistent entity-state tracking across the generation horizon.

Instruction-tuned models introduce an additional failure axis. Supervised fine-tuning (SFT) on dialogue data trains the model to replicate the statistical patterns of assistant responses; if the fine-tuning data is low-quality or misaligned with intended behavior, the model can learn to produce responses that superficially resemble instructions without executing them. In the extreme case evaluated here, the pre-patch chat model produced single-clause non-responsive outputs to open-ended story generation prompts, indicating that the alignment fine-tuning had overridden, rather than supplemented, the base completion capability.

Data patching- fine-tuning a pre-trained model on a small, curated dataset targeting specific failure behaviors- has been explored in the context of safety alignment and preference learning. This work applies an analogous principle to generation quality, asking whether behaviorally targeted supervision can reshape local generation statistics without full retraining.

Methods

Models

Two variants of TinyStoriesLLM were evaluated. The base model is a decoder-only transformer trained for autoregressive story completion, with approximately 5 million parameters and a local self-attention window of 256 tokens. The chat model is an instruction-tuned derivative of the same backbone, formatted for user-assistant dialogue. Neither model was trained with reinforcement learning from human feedback. Both were operated in greedy decoding mode.

Failure Taxonomy and Adversarial Prompts

Three failure categories were operationalized:

Repetition prompts were designed to induce lexical or phrasal looping by providing a structurally repetitive stem; the canonical example was the prefix "The dog ran and ran and ran and," which biases the model toward continuation via the same predicate-reduplication pattern.

Coherence prompts embedded logical contradictions in the narrative premise, requiring the model to generate a consistent story despite an inherently contradictory setup (e.g., "a cat that lives underwater and on land").

Character confusion prompts introduced two to three named characters with distinct attribute bindings at the outset (e.g., "Anna liked red, Ben liked blue, Clara liked green,") then required the model to maintain those bindings across extended generation.

A total of 150 adversarial prompts were constructed- 50 per failure category- and applied across both model variants. The responses were manually labeled by a single annotator(Claude Opus 4.6 Extended Think) for the presence of failure along two evaluation dimensions.

Evaluation Dimensions

Two dimensions were assessed, each expressed as a binary success rate:

Dimension 1: Instruction following and story generation. A response was scored as a success on instruction following if the model produced a response that addressed the prompt content, and on story generation if the response constituted a multi-sentence narrative. Average output length in words was recorded as a continuous correlate.

Dimension 2: Story quality across failure categories. For each of the three failure types, a success was scored if the generated output did not exhibit the targeted failure mode. An overall quality rate was computed as the proportion of all responses judged successful across all three categories.

Patch Dataset Construction

Two patch datasets were curated by hand. The base model patch comprised 51 short stories exhibiting the target behaviors: lexically varied vocabulary with no repetitive filler phrases, logically self-consistent narratives, and single, clean story endings. The chat model patch comprised 40 user-assistant conversation pairs in which the assistant produced a full multi-sentence narrative in response to an open-ended story prompt. Patch stories were composed such that named characters maintained consistent attributes throughout (e.g., a story in which two characters are introduced, have a conflict, and are reconciled, with distinct speech and action patterns preserved).

Fine-Tuning Procedure

Each model was fine-tuned on its respective patch dataset using standard next-token prediction loss:

$$\mathcal{L} = -\sum_{t} \log p_\theta(x_t \mid x_{<t}),$$

summed over all tokens in the patch corpus.

Results

Dimension 1: Instruction Following and Story Generation

Figure 1 presents before-and-after comparisons for the base and chat models across instruction following rate, story generation rate, and average output length.

The base model exhibited no change on either binary metric: both instruction following and story generation remained at 100% before and after patching. Average output length declined marginally from approximately 150 words to approximately 140 words post-patch.

The chat model showed a qualitatively different pattern. Prior to patching, the chat model achieved 0% on both instruction following and story generation, producing outputs averaging 12 words in length (e.g., a 12-word non-responsive clause in reply to a prompt requesting a full story). After patching on 40 curated examples, instruction following improved to 74%, story generation to 64%, and average output length rose to approximately 100 words.

Dimension 2: Story Quality by Failure Category

Figure 2 presents success rates for each failure category before and after patching.

For the base model, pre-patch success rates were uniformly low: 6% for repetition, 4% for coherence, and 8% for character, yielding an overall rate of 6%. Post-patch rates rose to 24%, 14%, and 22% respectively, with an overall rate of 20%- a gain of 14 percentage points.

For the chat model, pre-patch success rates were 40% for repetition and 0% for both coherence and character, giving an overall rate of 0%. Post-patch rates were 54%, 22%, and 18% for repetition, coherence, and character respectively, with an overall rate of 14%.

Discussion

The results reveal a sharp asymmetry between the two model variants that reflects fundamentally different failure etiologies. The chat model's pre-patch behavior- 0% instruction following, 12-word average output length- is consistent with a model whose alignment fine-tuning had displaced the underlying completion capability: the base capability for story generation existed in the backbone but was suppressed by a mismatch in the instruction-following data. 40 curated conversation pairs were sufficient to restore this capability, producing a 74-percentage-point improvement in instruction following. This is a striking result: it demonstrates that behavioral alignment failures in small models can be corrected with minimal data when the failure is distributional rather than representational.

The base model's trajectory is the inverse. Its pre-patch instruction following and story generation rates were already at ceiling (100%), reflecting that story completion is the task the base model was trained for. The failure modes it exhibited- repetition, incoherence, character confusion- are not behavioral misalignment but architectural ceiling effects. A 5-million-parameter model with a

256-token attention window lacks the representational capacity to maintain entity-state bindings across long generation horizons or to suppress degenerate high-probability continuations reliably. The 14-point improvement in overall story quality post-patch is real but modest; qualitative examination shows that repetition shifted in character- the post-patch base model produced less lexical looping but more narrative incoherence (e.g., a story about a dog that transitions from running in a park to being trapped in a tree hole with no causal connector). This pattern suggests that the patch data suppressed the specific high-frequency tokens associated with repetition but did not address the underlying generation dynamics.

Comparison across failure categories is instructive. Coherence remains the most refractory failure for both models: post-patch success rates for the base model (14%) and chat model (22%) are the lowest or near-lowest of all categories. Coherence violations require maintaining propositional consistency across multiple sentences, which depends on effective long-range attention—the precise capability that the 256-token window constrains. Character confusion is partially addressable by patch data for the base model (8% to 22%), suggesting that short narratives with consistent character naming can bias the model toward more stable entity tracking within its effective context window, even if this does not generalize to longer generation sequences.

The study's primary limitation is statistical power. With 30 evaluation prompts per model, each binary success rate carries substantial uncertainty; approximate 95% binomial confidence intervals at $n = 30$ span ± 18 percentage points at rates near 0.5, meaning that differences smaller than roughly 20 percentage points cannot be reliably distinguished from noise. A second limitation is the single-annotator labeling scheme: without inter-annotator agreement, the failure labels carry unknown measurement error that is not quantified.

Error Analysis

Measurement Error

The primary source of measurement error is the binary success/failure annotation scheme applied by a single annotator. Failure modes such as coherence and character confusion exist on a continuum- a story may partially maintain character attributes or be locally consistent but globally contradictory- and the threshold for "success" was not operationalized with explicit rubrics. This introduces subjective variability that is unquantified and uncontrolled. The absence of inter-annotator agreement statistics means that the reported success rates cannot be treated as objective measurements; they are better understood as informed judgments from a single evaluator. This constitutes systematic measurement error, as any idiosyncratic annotator tendency (e.g., leniency toward character confusion failures) would bias all rates in the same direction.

Dataset and Sample Size Error

The evaluation set of 150 prompts per model is insufficient for precise estimation of category-level success rates. The prompt set was adversarially constructed by the same investigator who built the patch dataset, creating a risk of inadvertent overfitting: if the adversarial prompts and the patch stories share surface-level features, the fine-tuned model may recover on the evaluation set via memorization rather than generalized behavior change.

The patch datasets are also small. The base model patch (51 examples) and chat model patch (40 examples) are within the regime where catastrophic forgetting of pre-training statistics is a concern. The observed decline in average output length for the base model (150 to 140 words) may reflect partial overwriting of the pre-trained length distribution, which would constitute a systematic side effect of the patching procedure.

Model and Implementation Error

The base model's post-patch behavior, wherein repetition shifted rather than disappeared, is consistent with a specific error mode: the patch data suppressed high-frequency repetitive tokens at training time, but the underlying probability mass was redistributed to other degenerate continuations rather than to coherent narrative development. This is a form of distributional displacement rather than correction, and its detection requires evaluation against held-out prompt types not targeted by the patch-which the current design does not include.

Mitigations

Several concrete mitigations would strengthen a follow-on study. Inter-annotator agreement (Cohen's $\kappa \geq 0.7$) should be established before reporting success rates. Evaluation set size should be increased to at least $n = 50$ per category per model to reduce binomial uncertainty to ± 14 points at $p = 0.5$. Patch examples should be drawn from a different distribution than the adversarial prompts to decouple construction bias. To test for memorization versus generalization, a withheld set of prompts structurally similar but lexically distinct from both training and evaluation prompts should be evaluated post-patch. Finally, reporting standard error on the success rate differences, and testing the null hypothesis of no change via a McNemar test on matched pre/post pairs, would allow statistical conclusions to be drawn about the significance of the observed improvements.